



Artificial Intelligence for Digital Characters

Metrics & Applications

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Evaluation of TTS

Responsible Systems
Applications
Exam

Intelligibility Test
Diagnostic Rhyme Test
Modified Rhyme Test

Recap: Word Error Rate

How to evaluate the word string output by a speech recognizer?

$$\text{Word Error Rate} = \frac{100 * (\text{Insertions} + \text{Substitutions} + \text{Deletions})}{\text{Total Word in Correct Transcript}}$$

REF: portable form of stores last night so

SYS: portable phone upstairs last night so

- Evaluation of TTS generally requires humans!
 - Listen to example utterances, rate various aspects (naturalness, intelligibility, friendliness, expressiveness, etc.). Scale of 1-5
 - Mean opinion score (MOS). Average of ratings
 - AB Tests (prefer A, prefer B)
- Intelligibility Tests
 - Did the human hear the correct thing?

- Diagnostic Rhyme Test (DRT)
 - Humans listen identification choice between two words differing by a single phonetic feature
 - 96 rhyming pairs
 - Veal/feel, meat/beat, vee/bee, zee/thee, etc
 - Subject hears “veal”, chooses either “veal” or “feel”
 - Subject also hears “feel”, chooses either “veal” or “feel”
 - Often embedded in carrier phrases (now we will say <word> again)
 - % of right answers is intelligibility score
- Modified Rhyme Test (MRT)
 - 50 sets of 6 words
 - Example set: went, sent, bent, dent, tent, rent
 - Listeners hear single word and must choose one word in set
- Reading addresses out loud, reading lines of news text, etc.

Evaluation of TTS

Responsible Systems

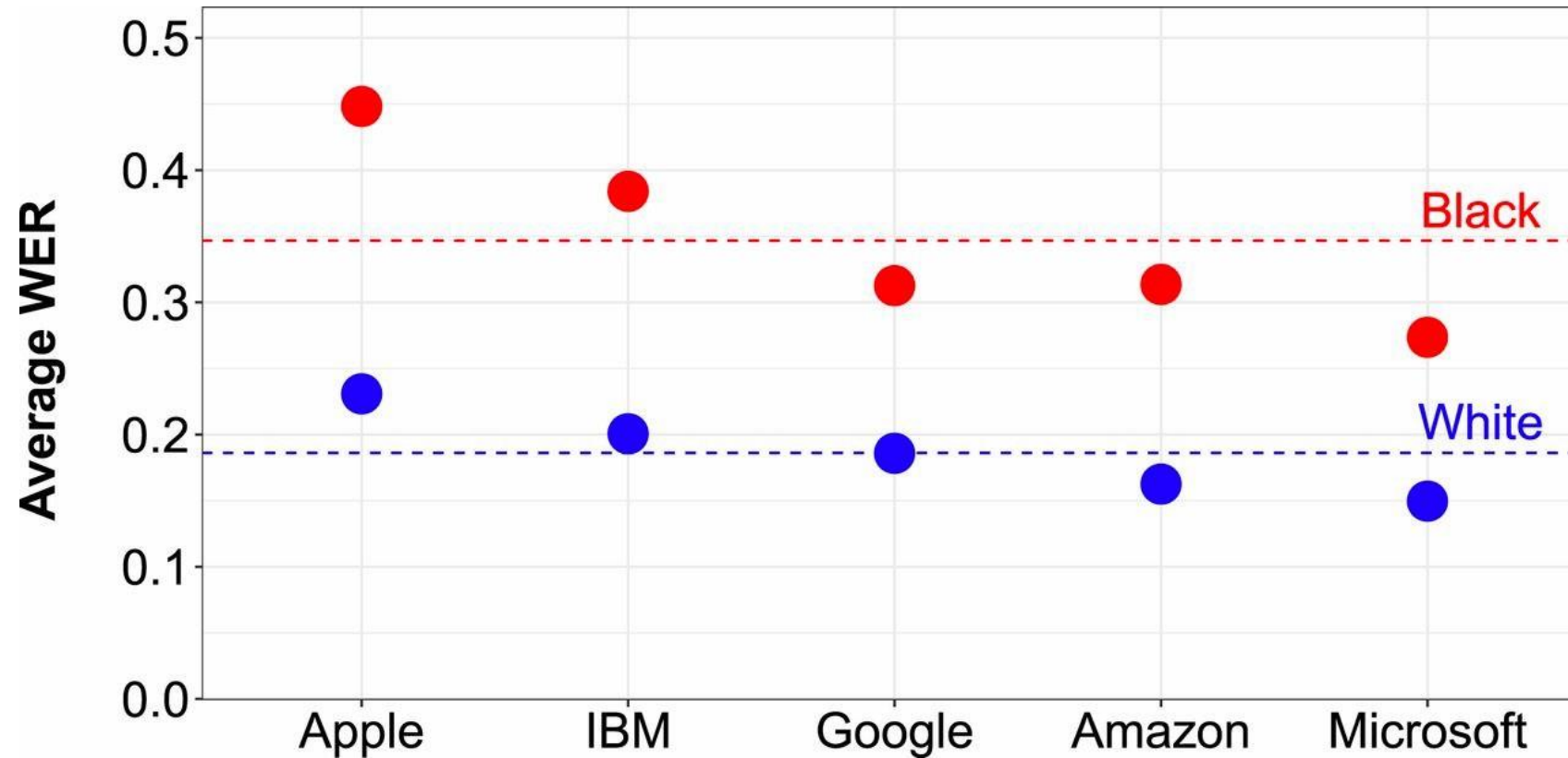
Applications
Exam

Bias
System Design Issues
Insufficiently Training
Distribution Shift

Broad categorization for this discussion:

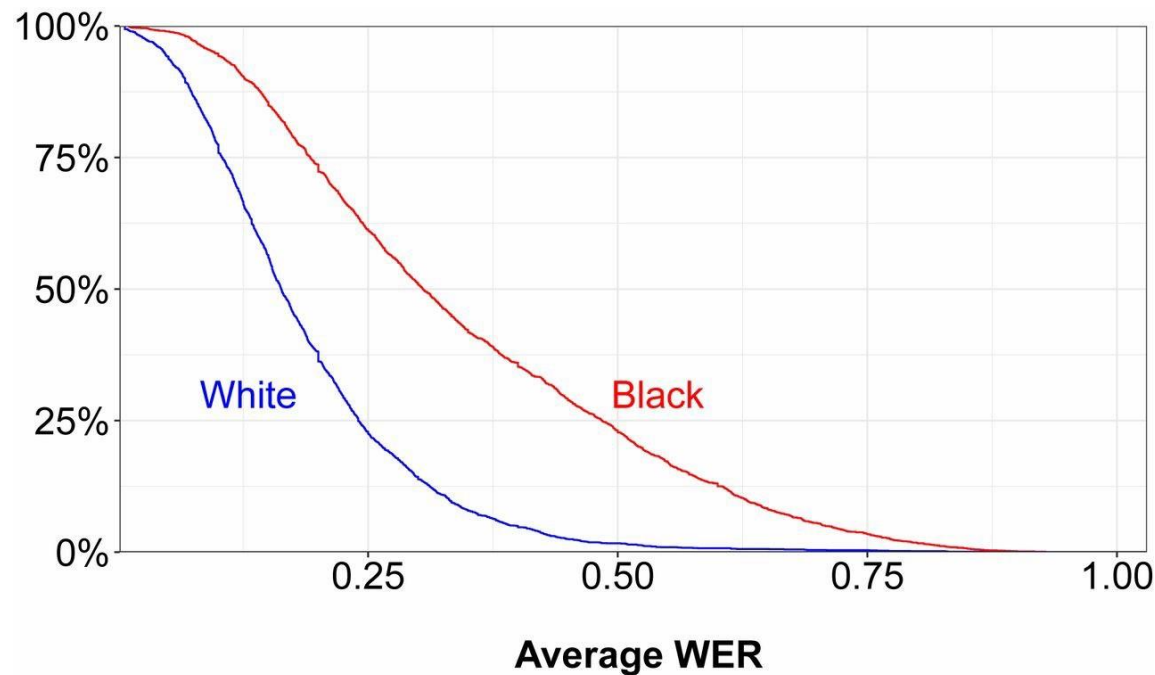
- Harmful system: negative impact on people or the world
- Biased system: performs differently, or not at all, for some subpopulation of inputs

Bias in ASR: Racial Disparities



(Koenecke *et al.* 2020)

23% of audio snippets from black speakers have WER > 0.5.
Compared to 1.6% of audio snippets from white speakers

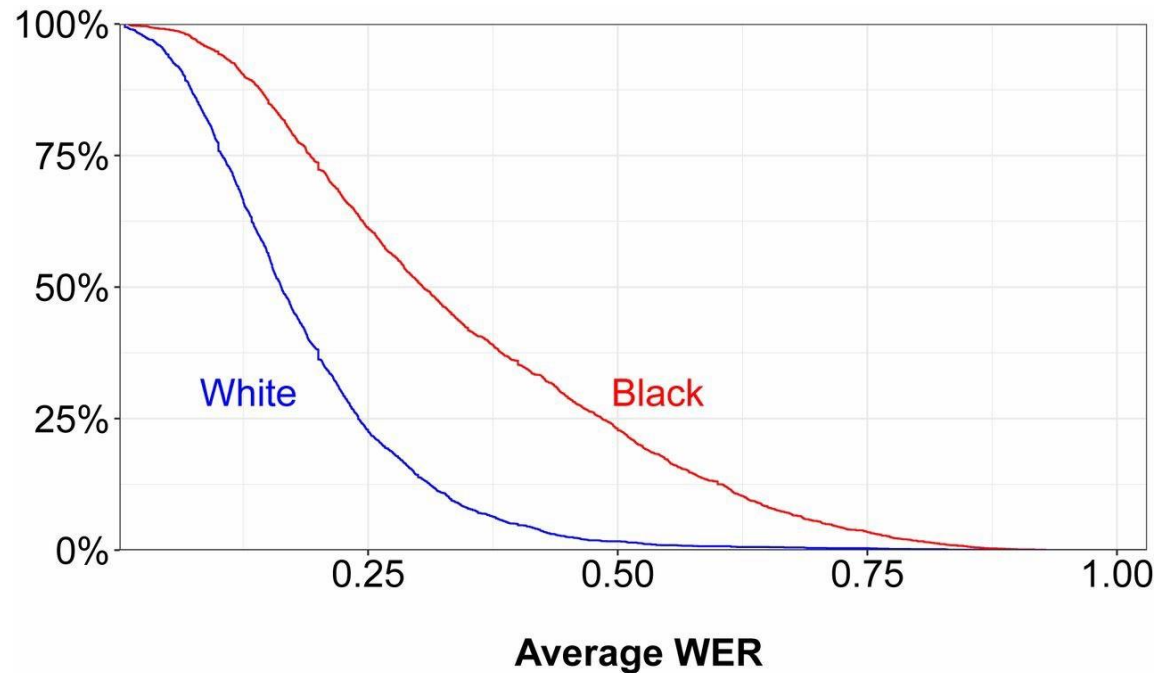


(Koenecke *et al.* 2020)

Bias in ASR: Racial Disparities

Causes? Differences attributed to acoustic model quality

- Training data distribution bias?
- Dialectal and accentual variations
- Speaker characteristics



(Koenecke *et al.* 2020)

What causes negative impacts of ML?
(think beyond just spoken language systems here)

- System design issues
- Insufficiently trained / low performing ML models
- Distribution shift and mis-application

- Problem formulation / ML task should be ethical
- Example:
 - Predict future criminality from face image
 - In speech, emotion recognition is sometimes controversial
 - Difficult to categorize the true range of emotions, and how individuals express them in speech
 - Need to be careful of how recognized emotions are used in a broader system.

- ML is never perfect. Potential negative impacts of mistakes
 - Lower quality experience for some inputs/users
- Often ML mistakes are not uniform random
 - Performance issues might disproportionately affect certain subgroups of input examples
 - Spoken language inputs are often correlated with demographic or medical factors affecting a person's voice

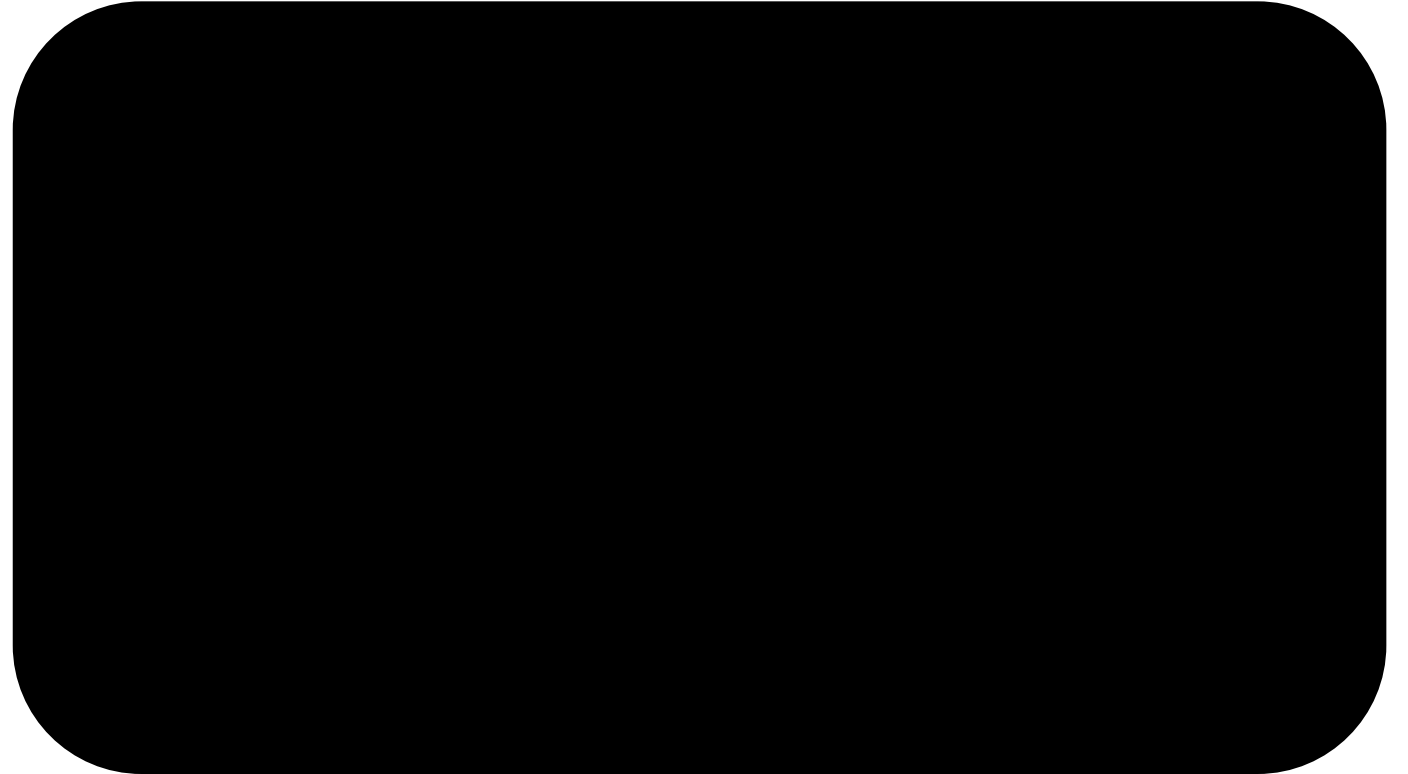
- Define and announce the expectations for a system
- Fundamentally comes back to (1) training data distribution and (2) generalization performance of models
 - Ensure training data is representative
 - Develop “unit test” evaluation sets to ensure sub-group performance is adequate

- Be aware of local laws if collecting audio data
- Generally data should be collected for a purpose
- Participants should be aware of any secondary uses of the data
 - GDPR makes this secondary use requirement more explicit

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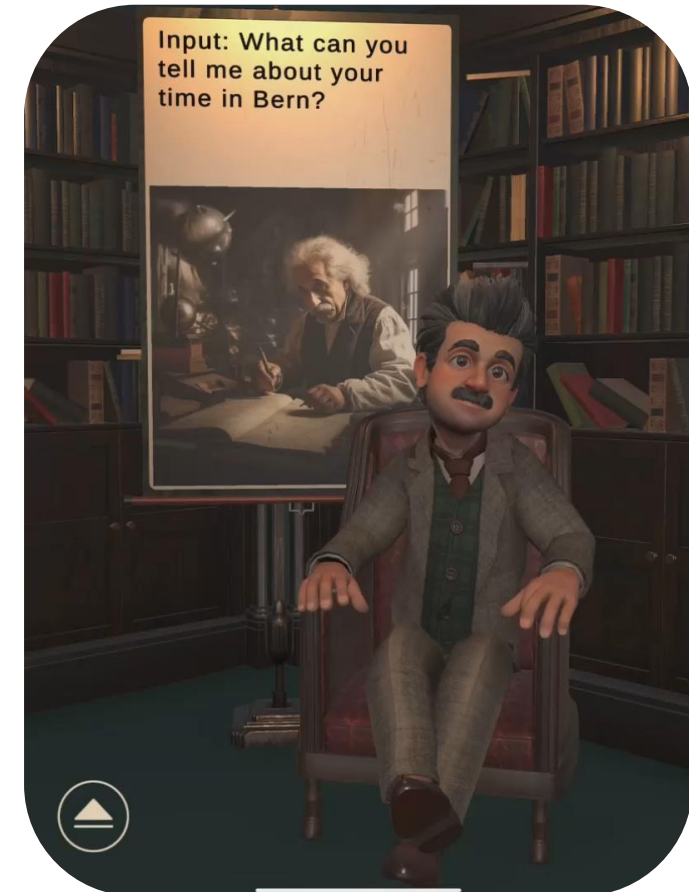
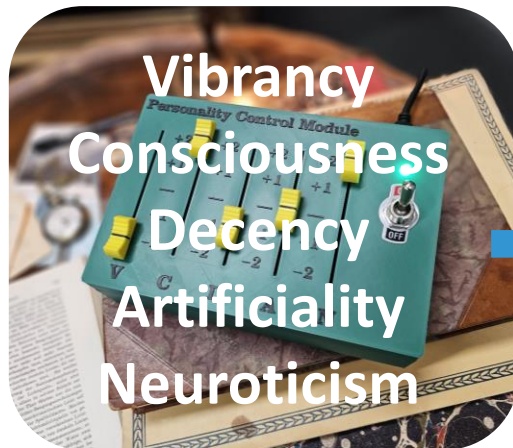
Digital Einstein
Virtual Doctor
Virtual Psychotherapist

- Response generation
 - GPT-4.1 mini (prompt engineering)
 - Llama 3 (fine-tuned, legacy)
- Speech synthesis
 - Microsoft Azure
 - Elevenlabs
- Animation synthesis
 - Salsa / Audio2Face (Face)
 - Motion-captured (Body)
- Automatic image generation
 - GPT Image
- Automatic sound generation
 - Elevenlabs
- Webcam data
 - User Characteristics
 - User Behavior
 - Reidentification

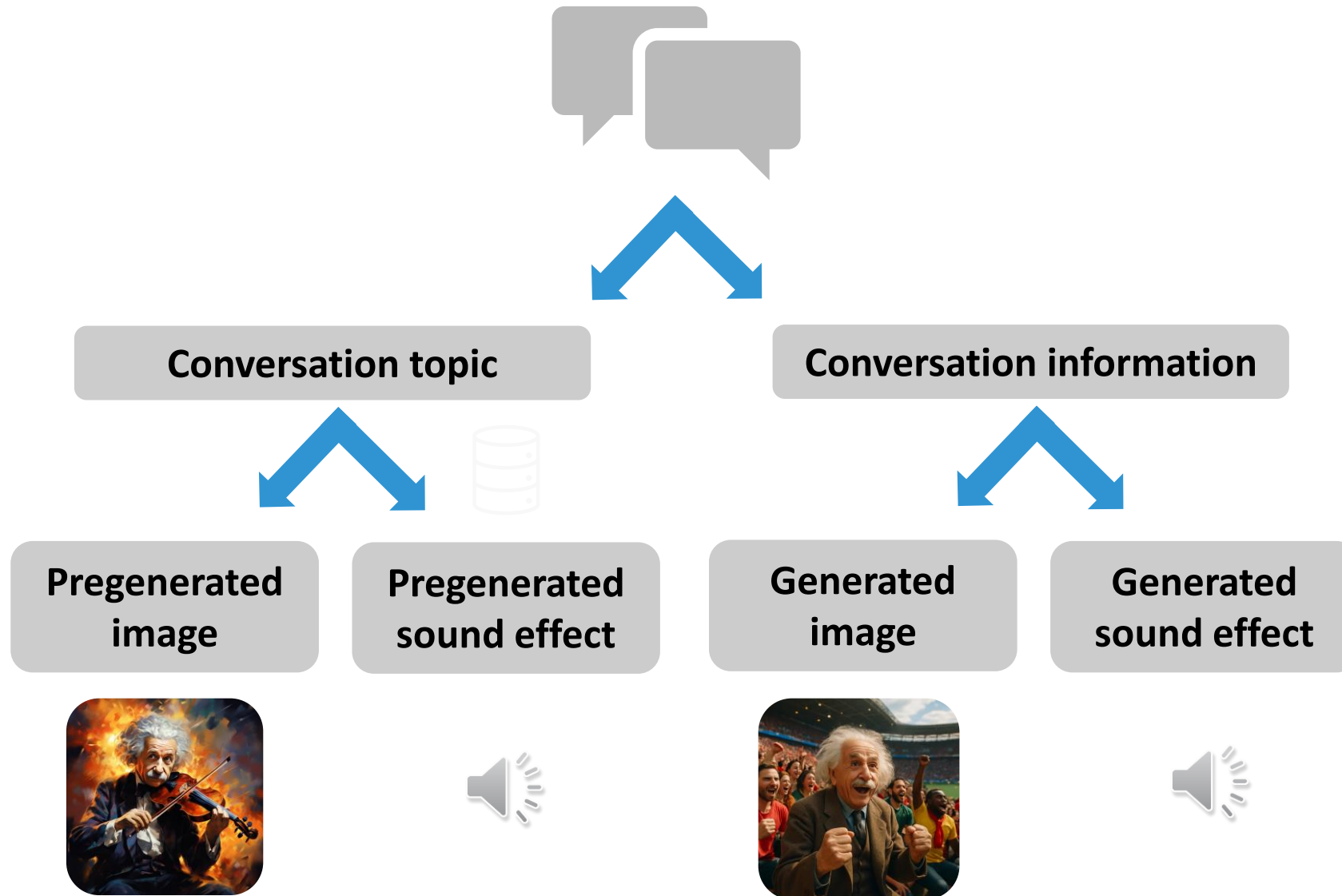




Digital Einstein – Personality Infusion



Digital Einstein – Storytelling Enhancements



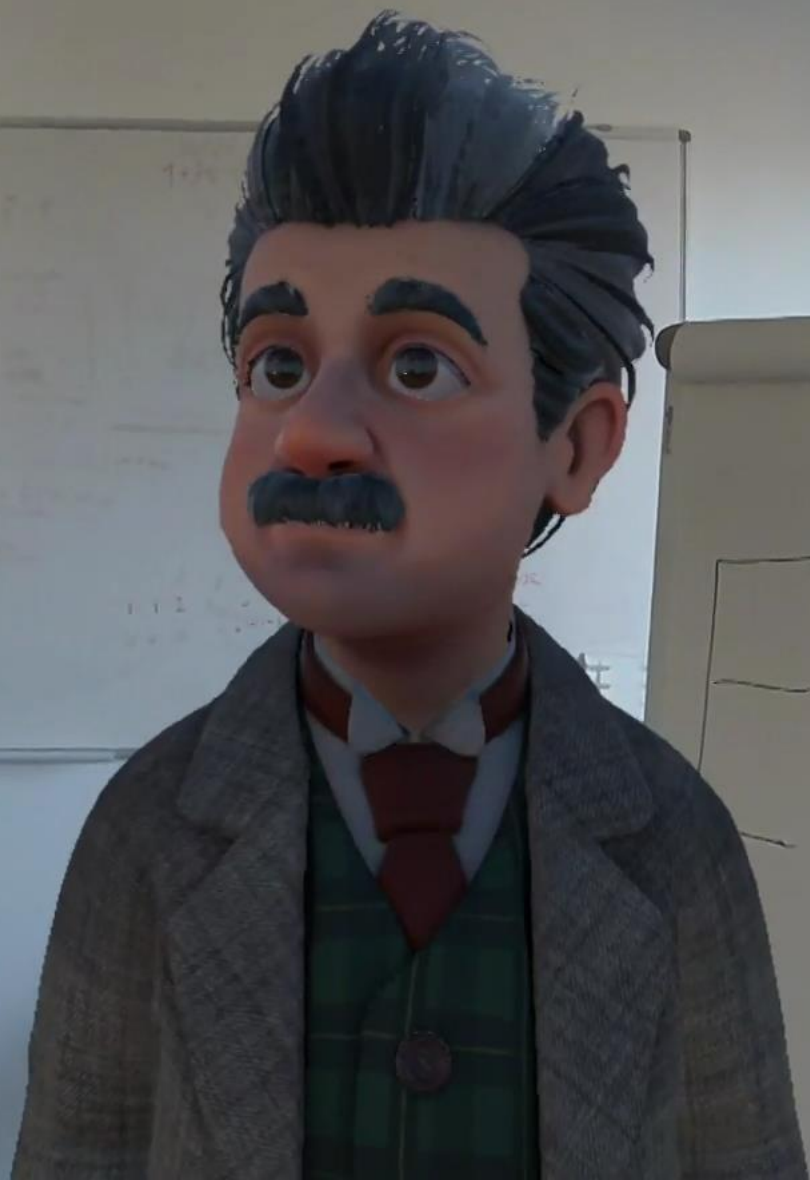
Digital Einstein – Body Animations

Old body animations



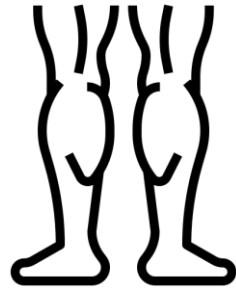
New body animations



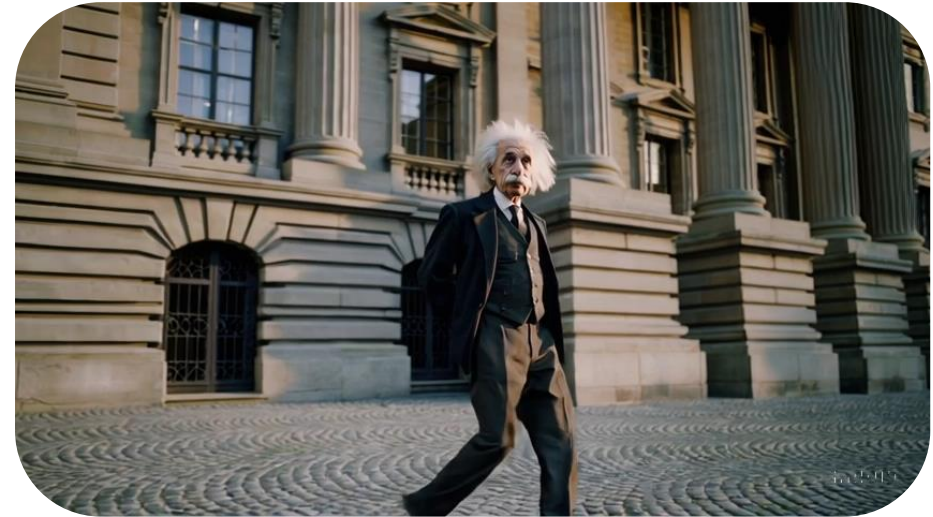


$$E = mc^2$$





Leg & feet animations



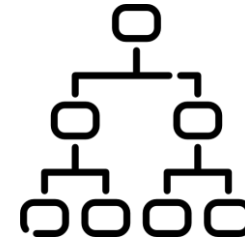
Videos instead of images



Digital Einstein – Upcoming Features

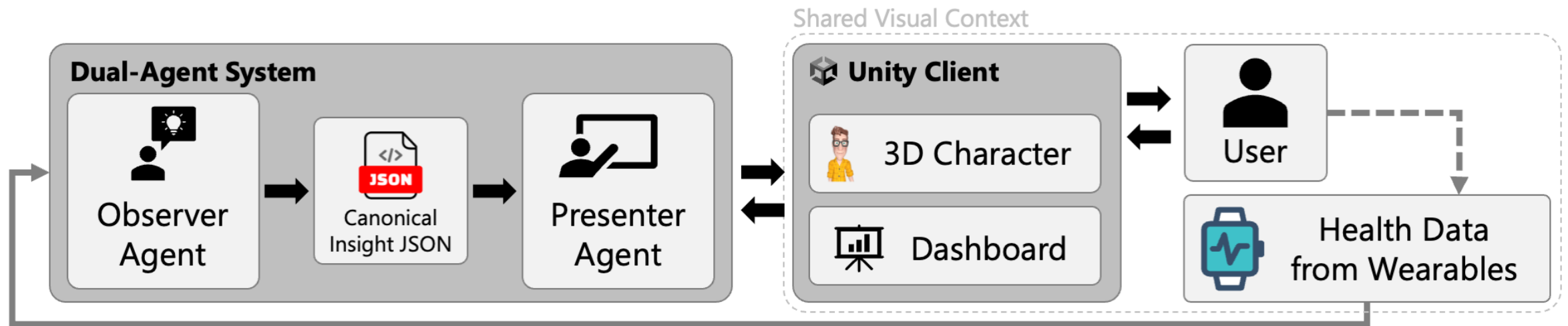


Runtime optimizations

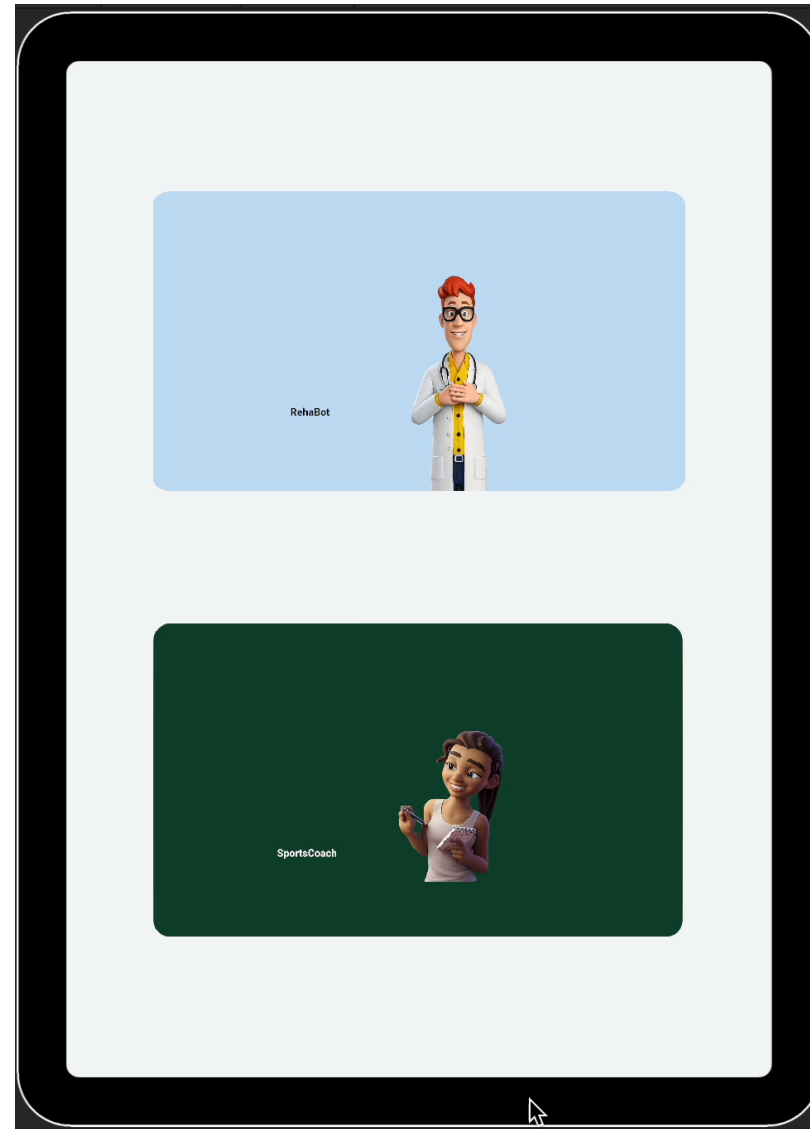


LLM & dialog tree fusion
Dynamic adaptation & personalization

Virtual Doctor – System Overview







Virtual Doctor – Experimental Setup

-  5 Participants
-  4 Configurations
-  2 Interactions per User
-  Questionnaires

Interaction Modes



1



2

Persona Profiles



A



B



Entry Q

Onboarding

Interaction 1

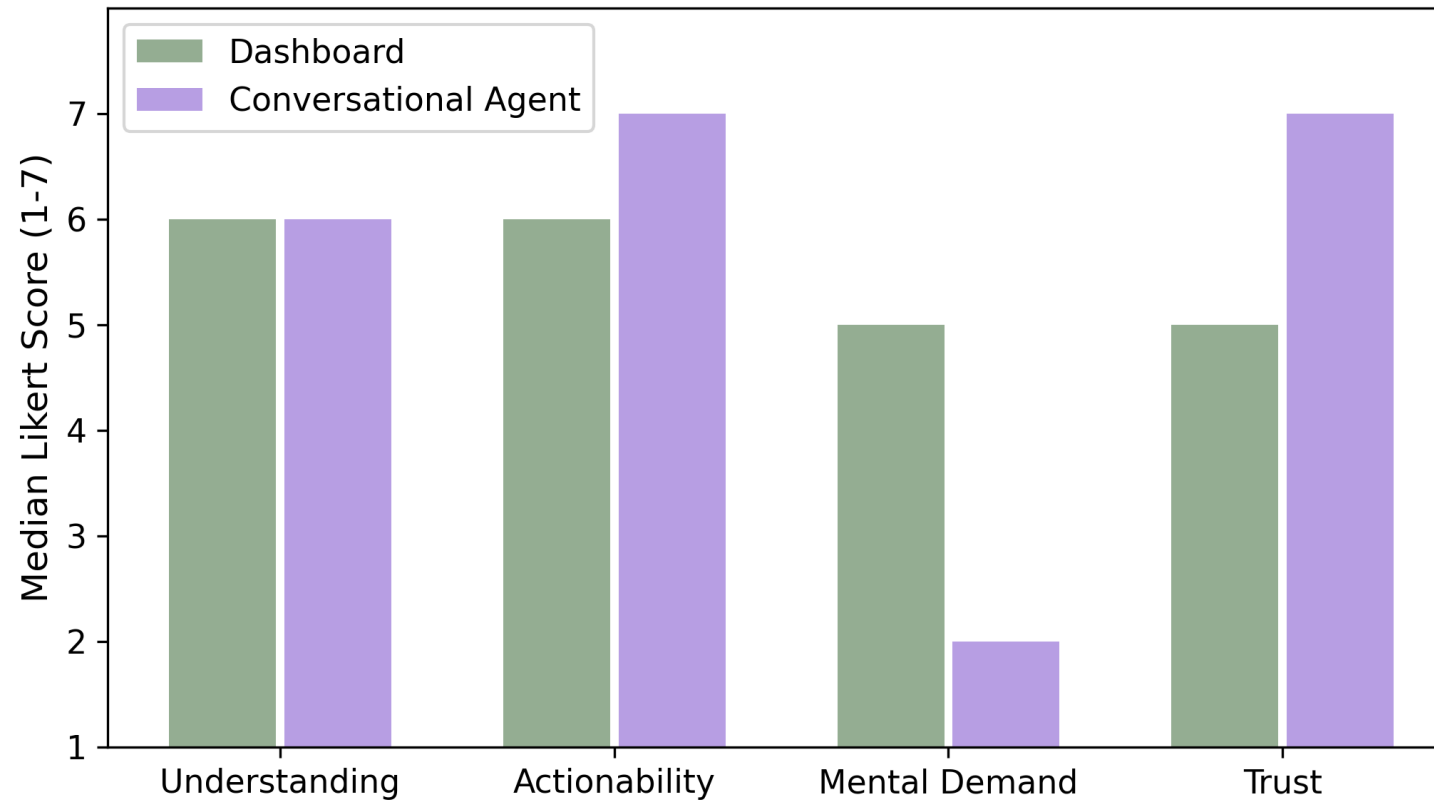
Post-Q 1

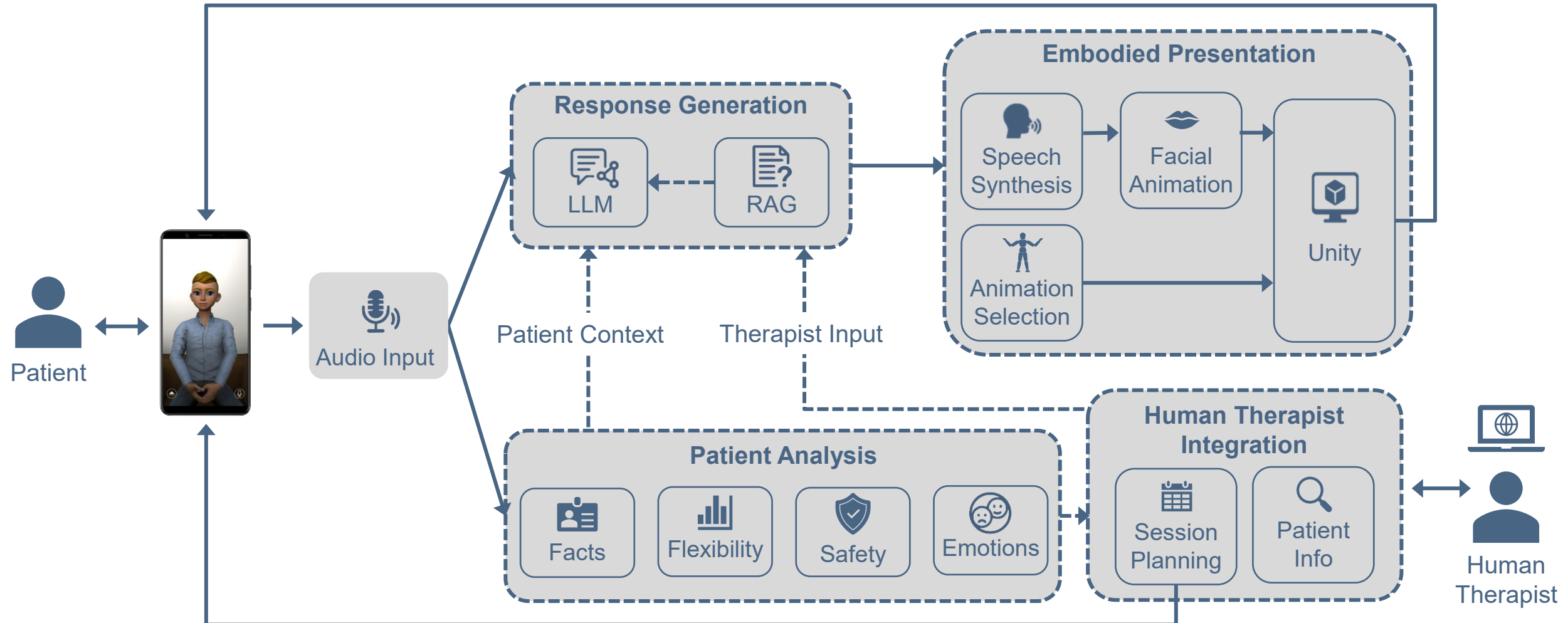
Interaction 2

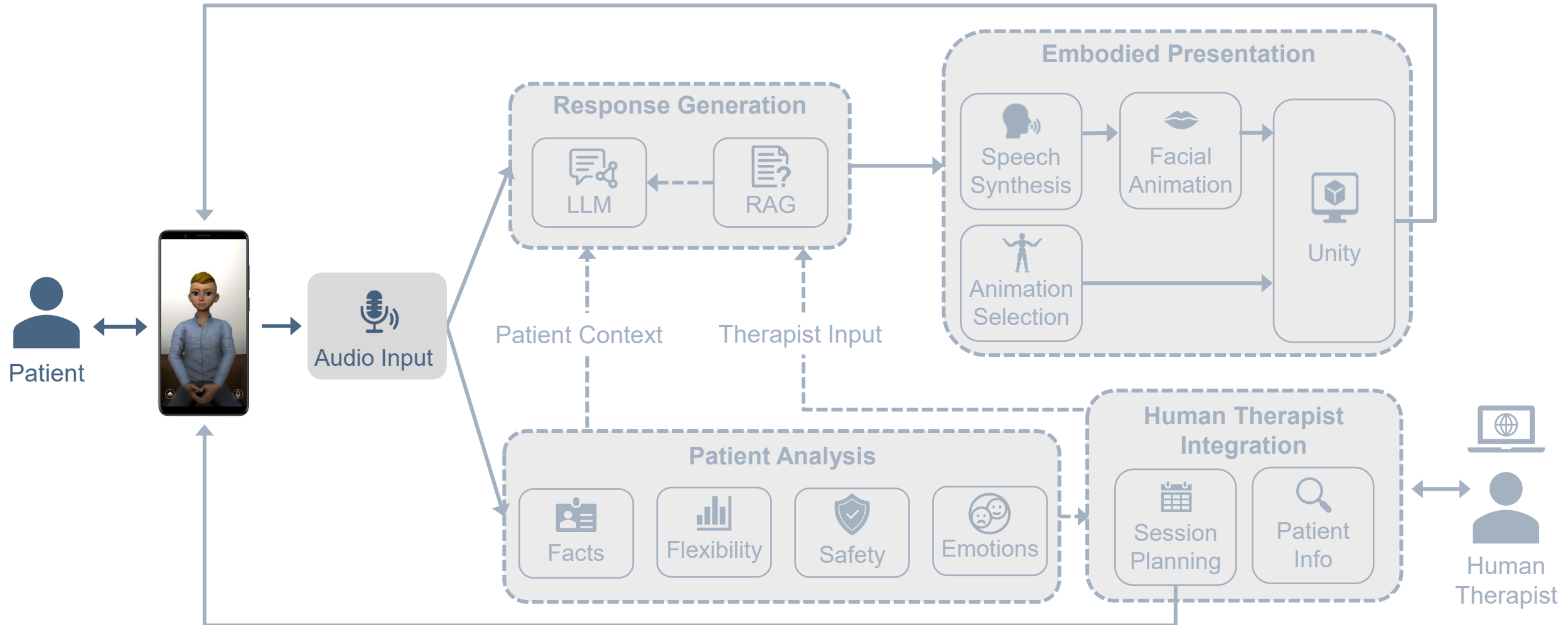
Post-Q 2

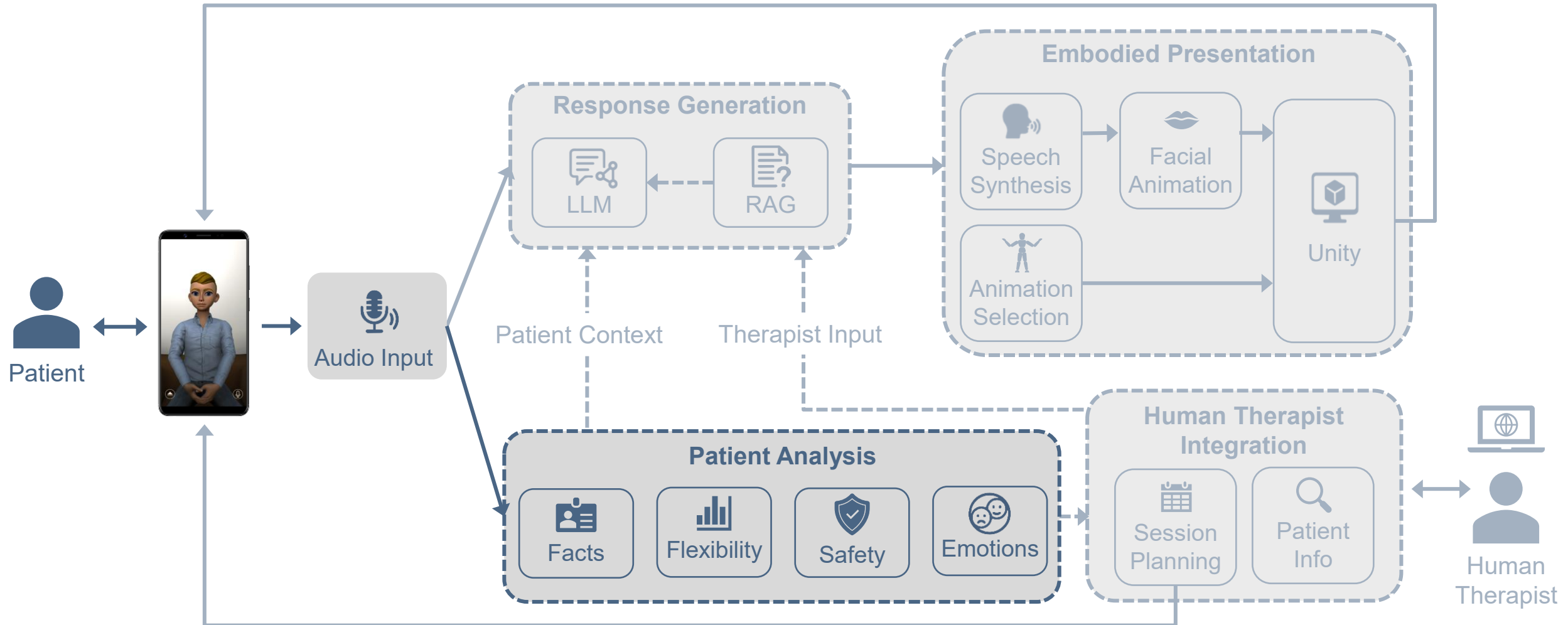
Exit Q

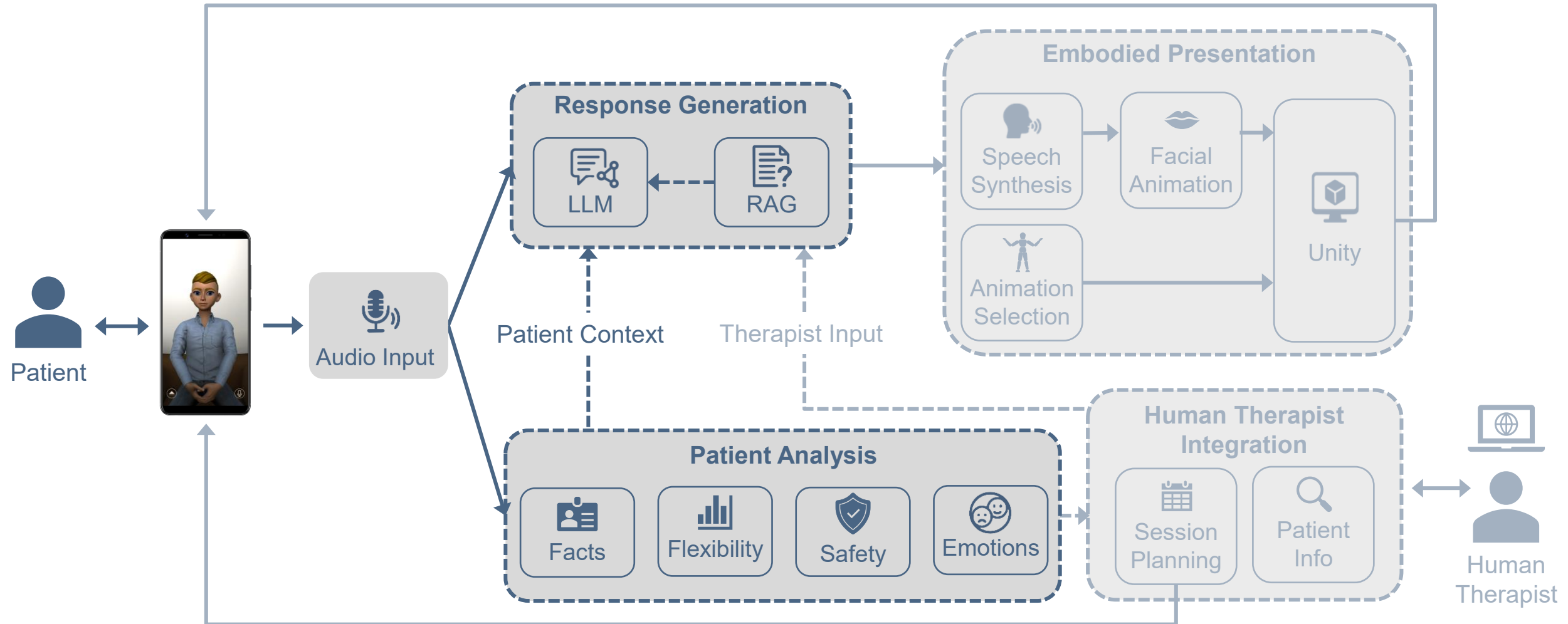
Virtual Doctor – Mode Preference

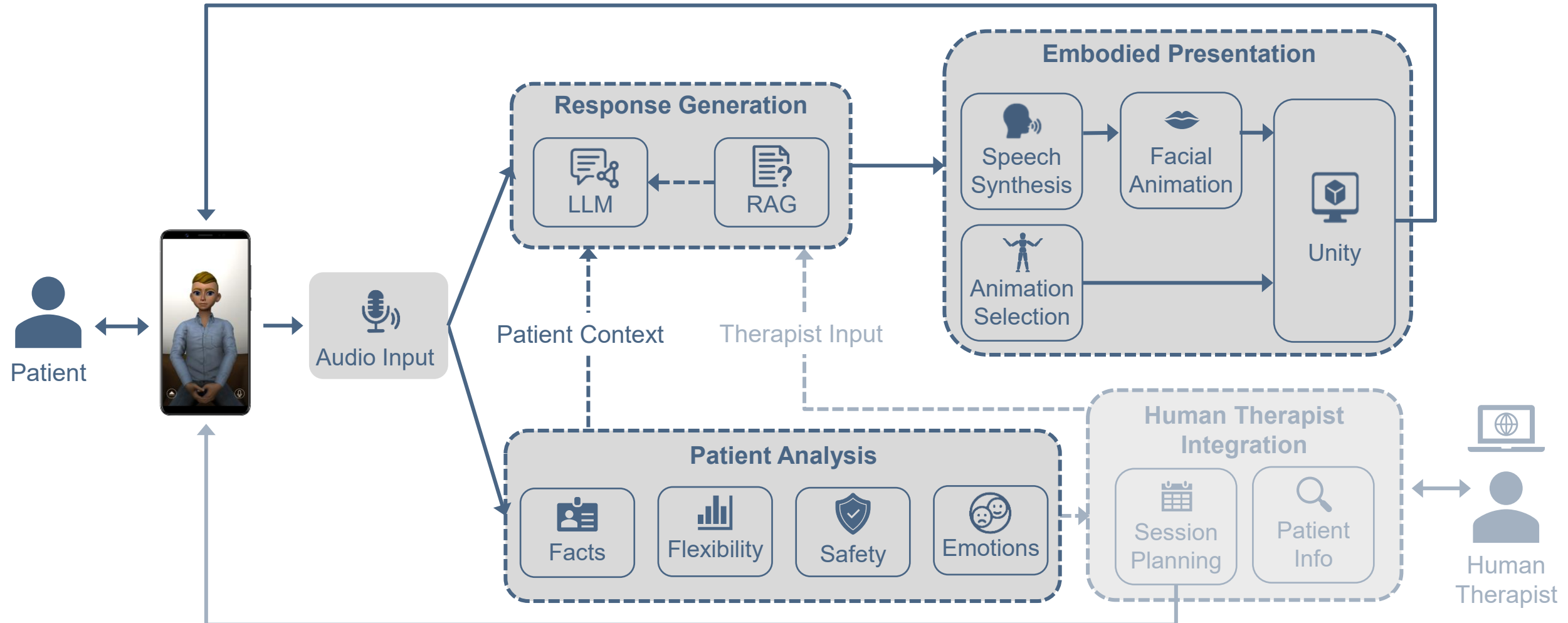


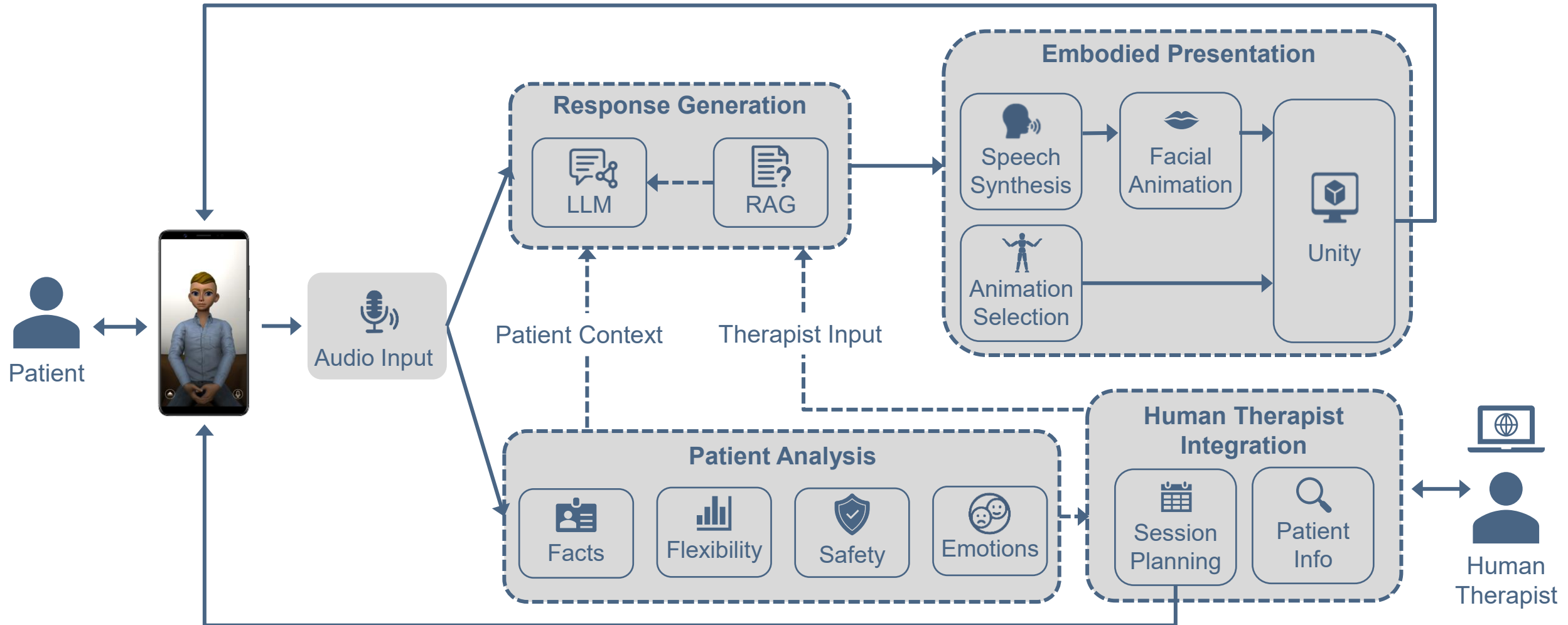




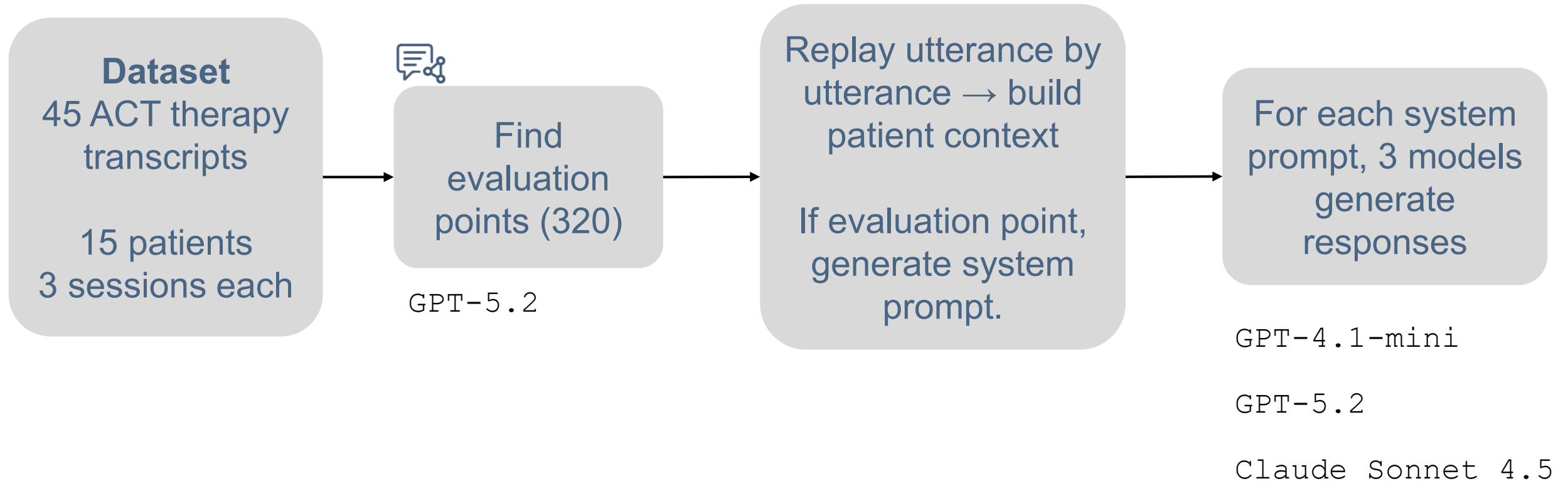


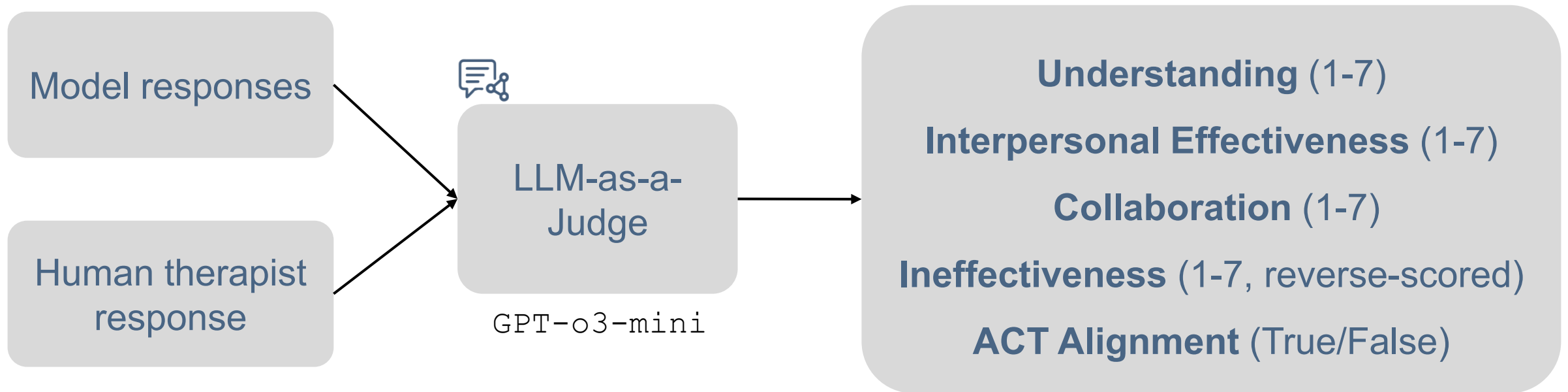












[Therapy Transcript Coding Manual](#)

Response Source	Understand. ↑	Interpers. ↑	Collab. ↑	Ineff. (rev.) ↑	Overall ↑	ACT Align. ↑
Human Therapist	4.27 (1.32)	4.28 (1.38)	3.87 (1.73)	6.26 (1.32)	4.67 (1.23)	0.65 (0.48)
GPT-4.1-mini	6.09 (0.70)	6.12 (0.69)	5.72 (1.38)	6.98 (0.12)	6.23 (0.63)	0.98 (0.15)
GPT-5.2	6.39 (0.83)	6.37 (0.83)	6.47 (0.92)	6.90 (0.58)	6.53 (0.71)	0.94 (0.23)
Claude Sonnet 4.5	6.47 (0.51)	6.45 (0.51)	5.97 (1.09)	6.99 (0.11)	6.47 (0.42)	0.98 (0.12)

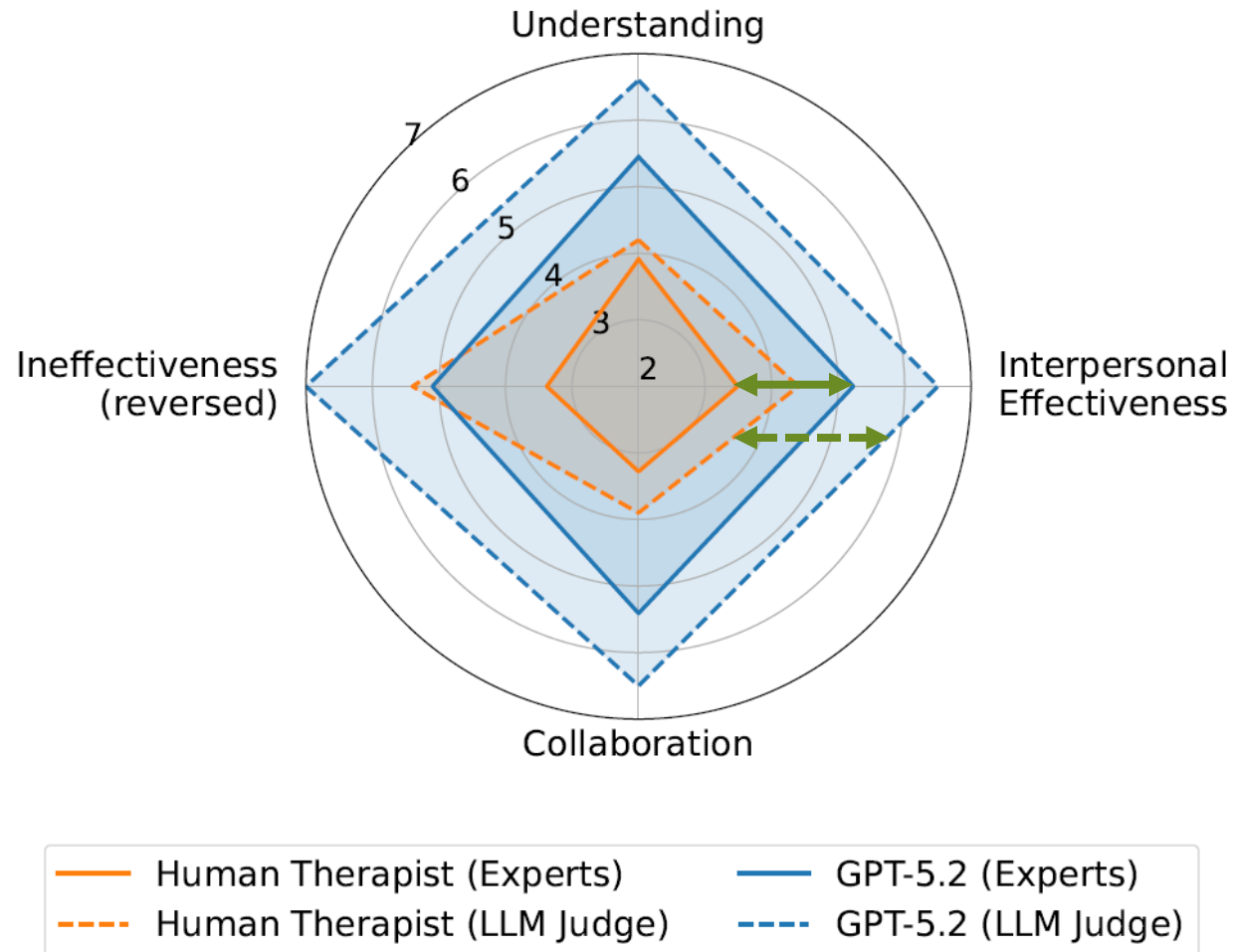
- Human therapist responses received **lowest ratings**
- GPT-5.2 and Claude Sonnet 4.5 outperformed GPT-4.1-mini
- GPT-5.2 achieved the **highest overall performance**

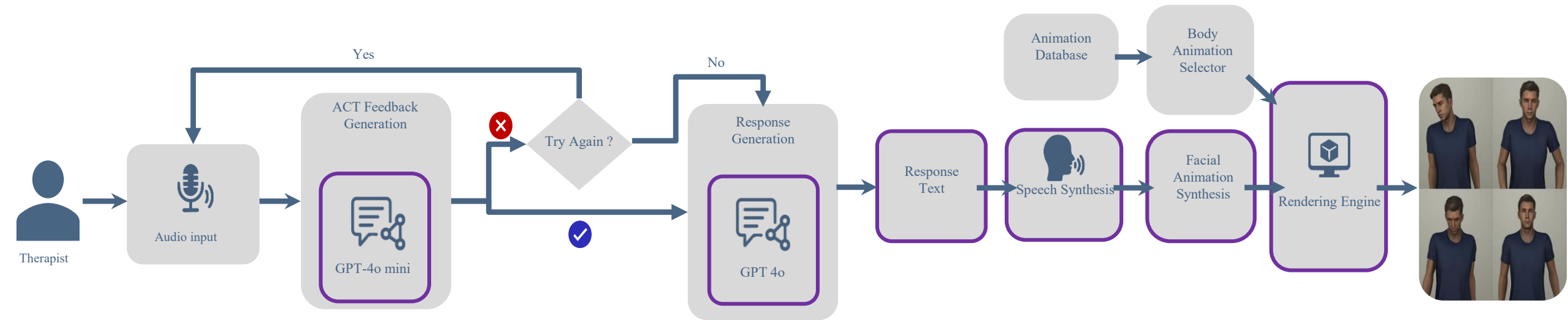
Participants

11 mental health professionals (10 female, 1 male)

Study Design

Each expert rated 10 response pairs (GPT-5.2 vs. human therapist), blinded to source



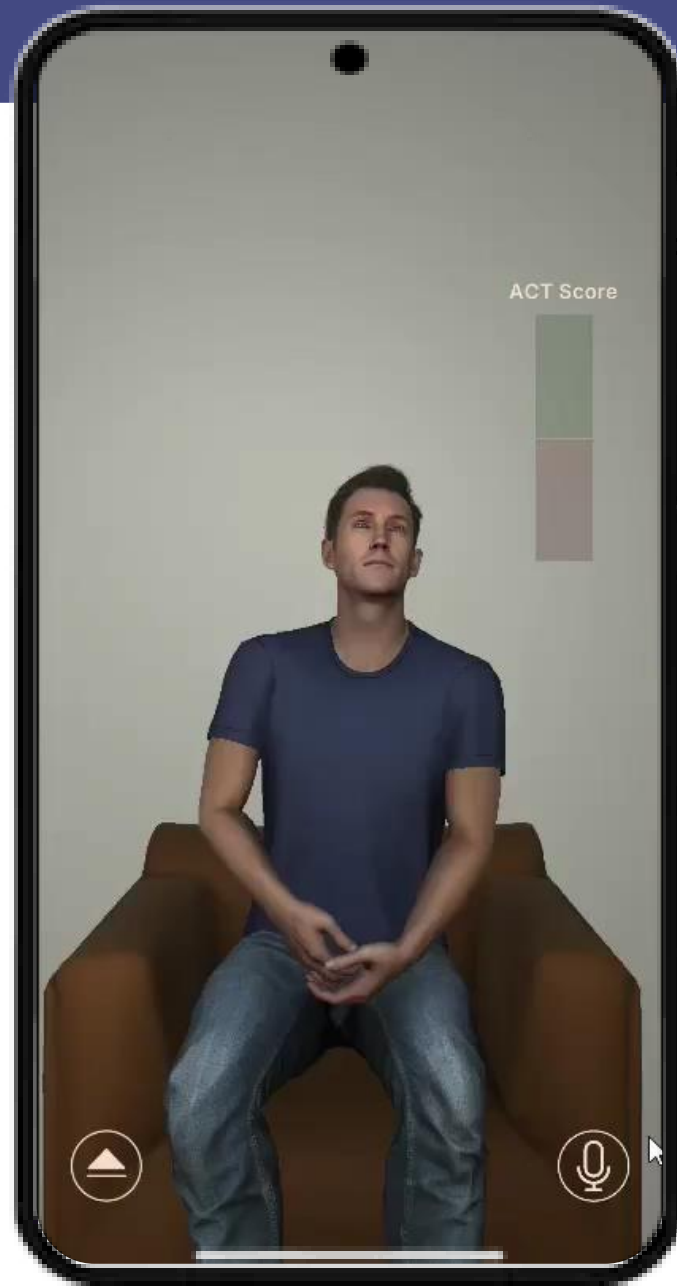


“What’s been feeling most overwhelming for you lately?”



“It’s mostly work that’s been overwhelming lately.”





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- Date
 - August, 17th
 - 12:00 to 14:00
 - room TBA
- No summary allowed at the exam
- Non-programmable calculator allowed
- Neutral dictionary (mother tongue to English) allowed
- Everything covered in the lecture and exercises relevant for the exam
- Tasks (120 points)
 - Multiple choice
 - Open questions

The end

